Locally Valid Dantzig–Wolfe Optimality Cuts

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Joint work with Guodong Lyu

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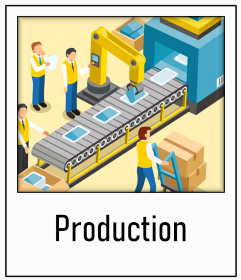


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Motivation

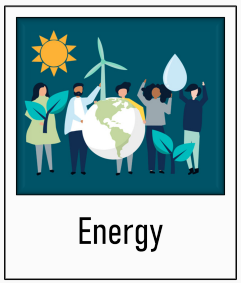
- Real-world planning problems are often dynamic, affected by uncertainty, and involve discrete decisions.

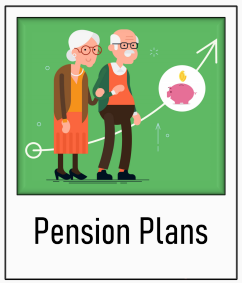

Finance


Production


Transportation


Logistics


Energy


Pension Plans

Motivation

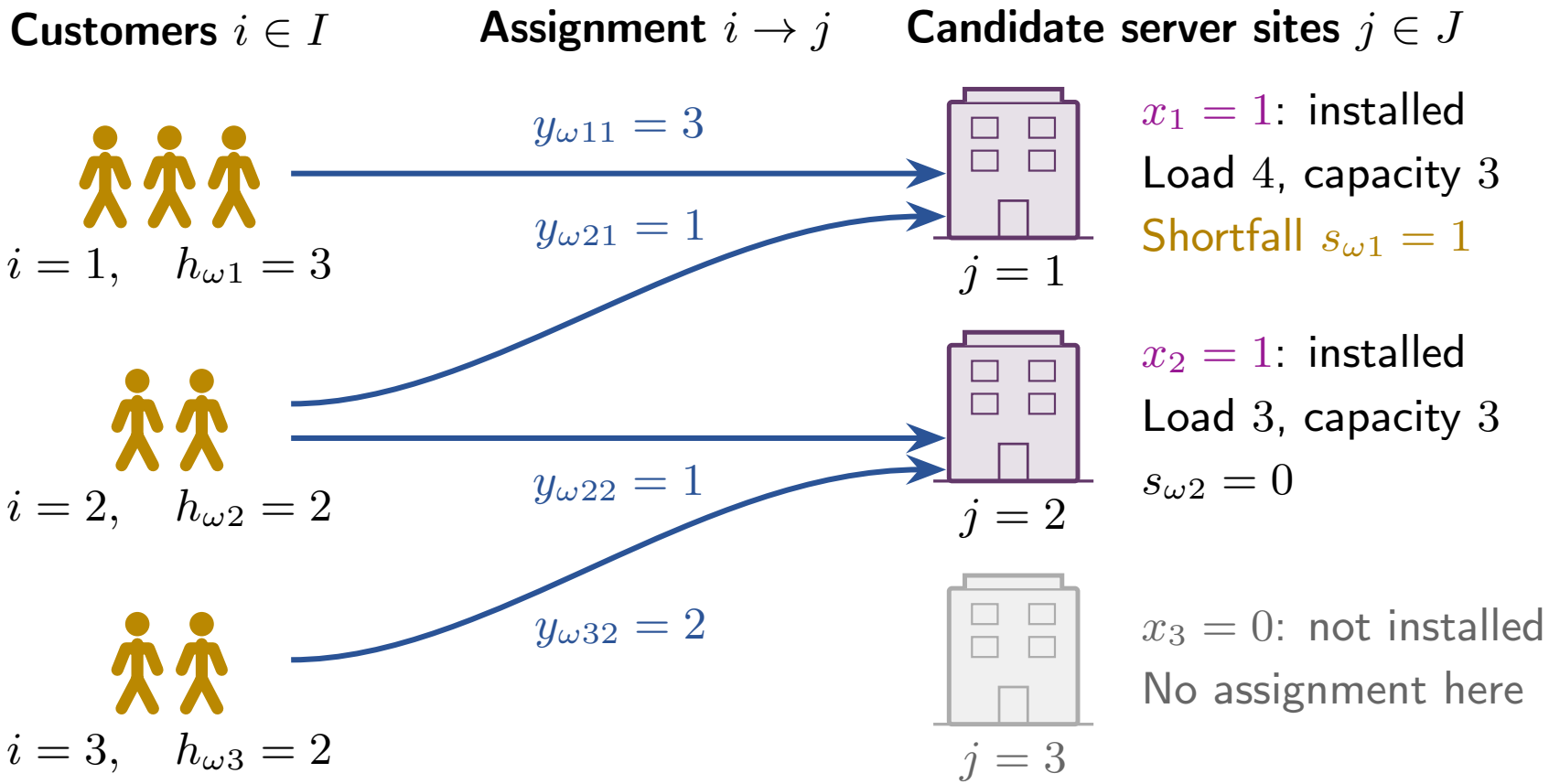
- Real-world planning problems are often dynamic, affected by uncertainty, and involve discrete decisions.



- A common structure is:
  - Decide a "here-and-now" plan first;
  - Take "wait-and-see" (recourse) actions after the scenario is realized.

Two-stage stochastic mixed-integer programming (TS-SMIP) models.

Stochastic Server Location and Sizing (SSLS)



SSLS: First-stage Problem

Stage 1: Choose installations x before demand is observed.

$$\min_{x} \quad \underbrace{\sum_{j \in J} c_j x_j}_{\text{installation cost}} + \underbrace{\sum_{\omega \in \Omega} p_\omega Q_\omega(x)}_{\text{expected recourse cost}}$$

$$\text{s.t.} \quad \sum_{j \in J} x_j \leq b$$

$$x_j \in \{0, 1\} \qquad j \in J.$$

- $x_j = 1$: install a server at site j, with cost c_j and capacity w.

SSLS: Recourse Value Function

Stage 2: Assign customers y after observe demands h.

- $y_{\omega ij}$: number of customers at location i assigned to site j.
- $s_{\omega j}$: capacity shortfall at site j, penalized per unit.

$$
\begin{aligned}
Q_\omega(x) := \min_{y_\omega, s_\omega} \quad & \underbrace{\sum_{j \in J} q_{0j} s_{\omega j}}_{\text{shortage penalty}} - \underbrace{\sum_{i \in I} \sum_{j \in J} q_{ij} y_{\omega ij}}_{\text{assignment revenue}} \\
\text{s.t.} \quad & \sum_{i \in I} d_{ij} y_{\omega ij} \le w x_j + s_{\omega j} && j \in J \\
& \sum_{j \in J} y_{\omega ij} = h_{\omega i} && i \in I \\
& y_{\omega ij} \in \{0, 1, \ldots, v\}, \quad s_{\omega j} \ge 0 && i \in I,\ j \in J
\end{aligned}
$$

- d_{ij}: capacity use per customer; q_{ij}: revenue per assigned customer.
- q_{0j}: shortage penalty per unit; v: maximum customers at each location.

Mathematical Formulation

We study the TS-SMIP problem:

$$z^\star = \min \quad c^\top x + \overbrace{\sum_{\omega \in \Omega} p_\omega Q_\omega(x)}^{\text{Expected recourse cost}}$$

$$\text{s.t.} \quad \boxed{\begin{aligned} &Ax \geq b \\ &x \in \mathbb{Z}^{n_1} \times \mathbb{R}^{n_2} \end{aligned}} \quad \rightsquigarrow \quad x \in X$$

Mathematical Formulation

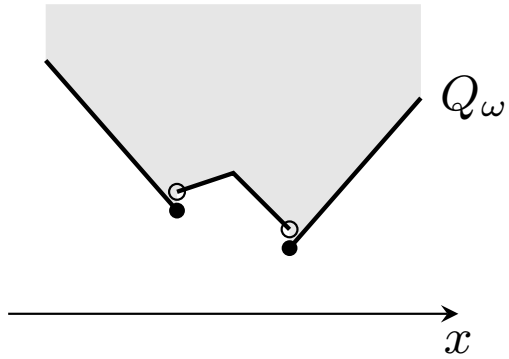
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where each scenario recourse problem is

$$\begin{aligned} Q_\omega(x) := \min \quad & q_\omega^\top y \\ \text{s.t.} \quad & T_\omega x + W_\omega y \geq h_\omega \\ & y \in \mathbb{Z}^{m_1} \times \mathbb{R}^{m_2} \end{aligned}$$



Mathematical Formulation

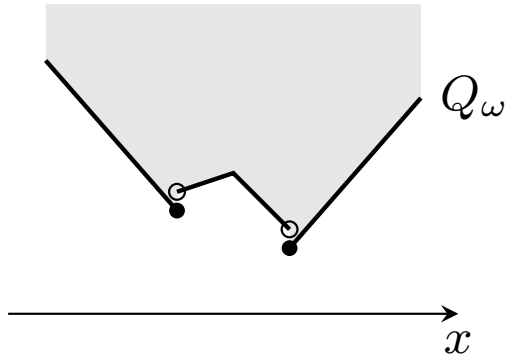
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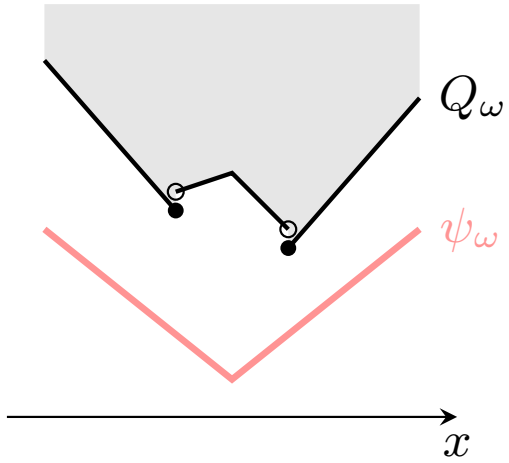


Issue: No closed-form expression of Q_ω

Value Function Approximation

Use a piecewise-linear convex approximation $\psi_\omega(x)$:

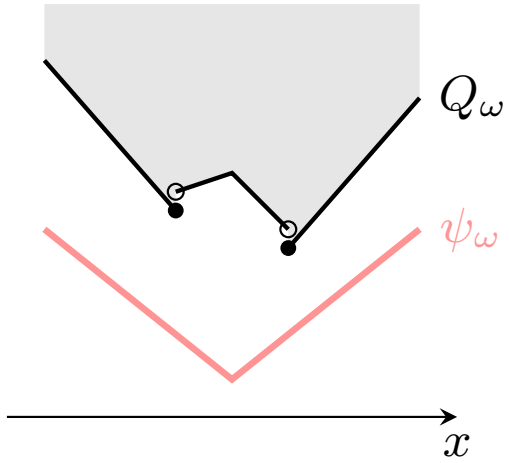
$$\psi_\omega(x) := \min \left\{ \theta_\omega \;\middle|\; \theta_\omega \geq v_\omega^\ell + {\lambda_\omega^\ell}^\top x \quad \forall \ell \in \mathcal{T}_\omega \right\}$$



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Master problem: Approximate value function Q_ω via $\psi_\omega(x)$

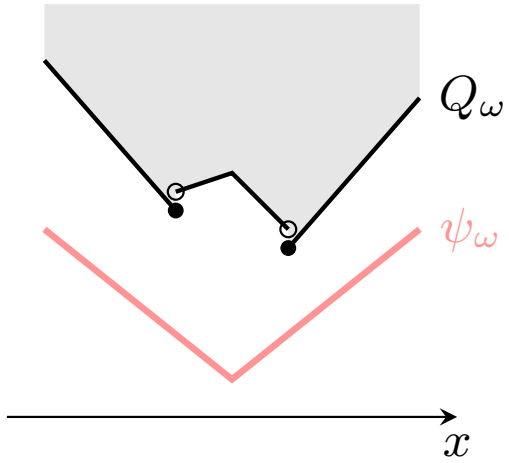
$$(MP) \qquad \min \quad c^\top x + \sum_{\omega \in \Omega} p_\omega ~~Q_\omega(x)~~ \psi_\omega(x)$$

$$\text{s.t.} \quad x \in X$$

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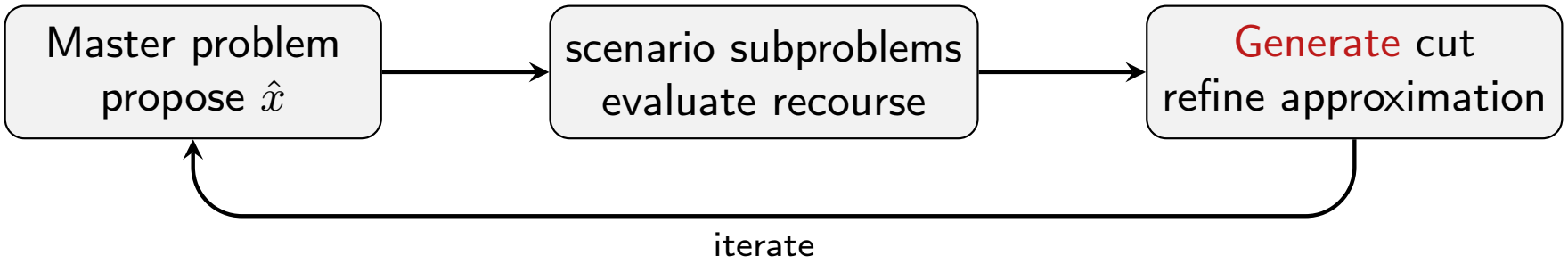
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Reformulation based on $\mathcal{Z}$

For cut generation, rewrite the scenario recourse problem as

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\begin{aligned}
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& z = x
\end{aligned} \tag{NA}
$$

- The region $\mathcal{Z}$ determines where the resulting cut must be valid.

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Benders cuts:	**Lagrangian cuts:**
Cheap but weak	Strong but expensive

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Benders cuts:	**Lagrangian cuts:**
Cheap but weak	Strong but expensive

We propose Dantzig–Wolfe cuts:
cheap & strong.

Dantzig–Wolfe Cuts

Dual LP + pricing + cut properties

Dantzig–Wolfe (DW) Dual

The DW reformulation is given as follows ($\Delta = \{\mu \geq 0 \mid \sum_j \mu_j = 1\}$)

$$\widehat{Q}_\omega(\hat{x}; \mathcal{Z}) = \min_{\mu \in \Delta} \quad \sum_{j=1}^{K_\omega} (q_\omega^\top y_\omega^j)\mu_j \tag{5a}$$

$$\text{s.t.} \quad \sum_{j=1}^{K_\omega} z_\omega^j \mu_j = \hat{x}. \tag{5b}$$

$\{(z_\omega^j, y_\omega^j) \mid j = 1, \ldots, K_\omega\}$: the extreme points of $\mathrm{conv}(\mathcal{S}_\omega(\mathcal{Z}))$.

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The LP dual of problem (5) gives the **DW dual** problem:

$$\max_{v_\omega, \lambda_\omega} \quad v_\omega + \lambda_\omega^\top \hat{x}$$

$$\text{s.t.} \quad v_\omega + \lambda_\omega^\top z_\omega^j \leq q_\omega^\top y_\omega^j \qquad \forall j \in \{1, \ldots, K_\omega\}.$$

Issue: Extreme points are unknown

DW Dual Relaxation

Use stored feasible points (z_ω^j, y_ω^j), indexed by $\mathcal{J}_\omega \subseteq \{1, \ldots, K_\omega\}$:

$$\begin{aligned} \text{(DWR)} \qquad \max \quad & v_\omega + \lambda_\omega^\top \hat{x} \\ \text{s.t.} \quad & v_\omega + \lambda_\omega^\top z_\omega^j \leq q_\omega^\top y_\omega^j \qquad \forall j \in \mathcal{J}_\omega. \end{aligned}$$

Check the missing inequalities by pricing.

$(\hat{v}_\omega, \hat{\lambda}_\omega)$: the optimal solution to (DWR)

$$\begin{aligned} \hat{\zeta}_\omega = \max \quad & \hat{v}_\omega + (\hat{\lambda}_\omega)^\top z - q_\omega^\top y \\ \text{s.t.} \quad & (z, y) \in \mathcal{S}_\omega(\mathcal{Z}) \end{aligned}$$

DW Dual Relaxation

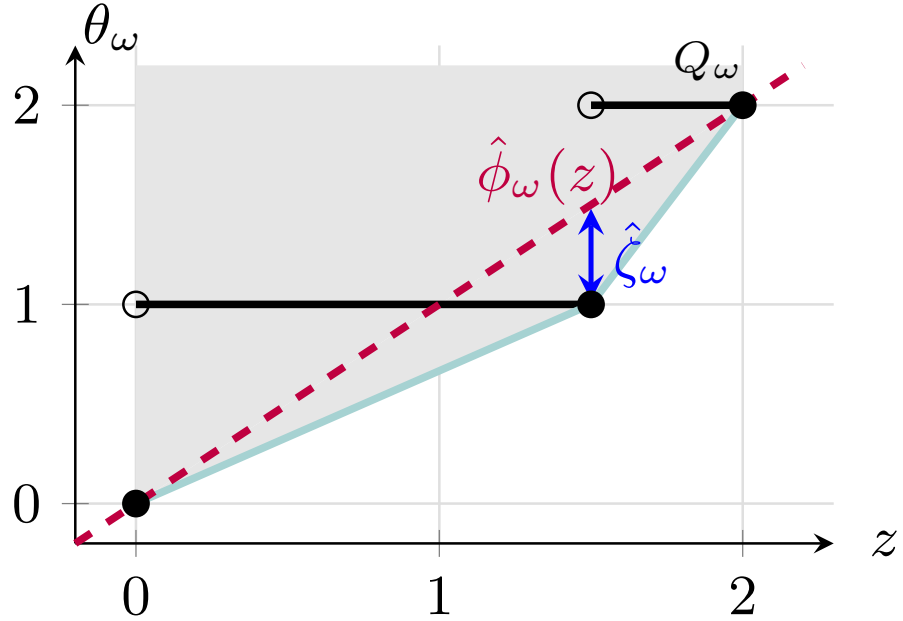
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Locally Valid Cuts

State screening + local recourse envelopes

Local Validity

- **Global validity.** A *globally valid* cut satisfies

$$v_\omega + \lambda_\omega^\top x \leq Q_\omega(x) \qquad \forall x \in X.$$

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Construction: enforce validity on a region $\mathcal{Z}$ satisfying

$$X^\star \subseteq \mathcal{Z} \subseteq X.$$

State Screening

Given $\psi_\omega \leq Q_\omega$ on $\mathcal{Z}$ and an upper bound UB:

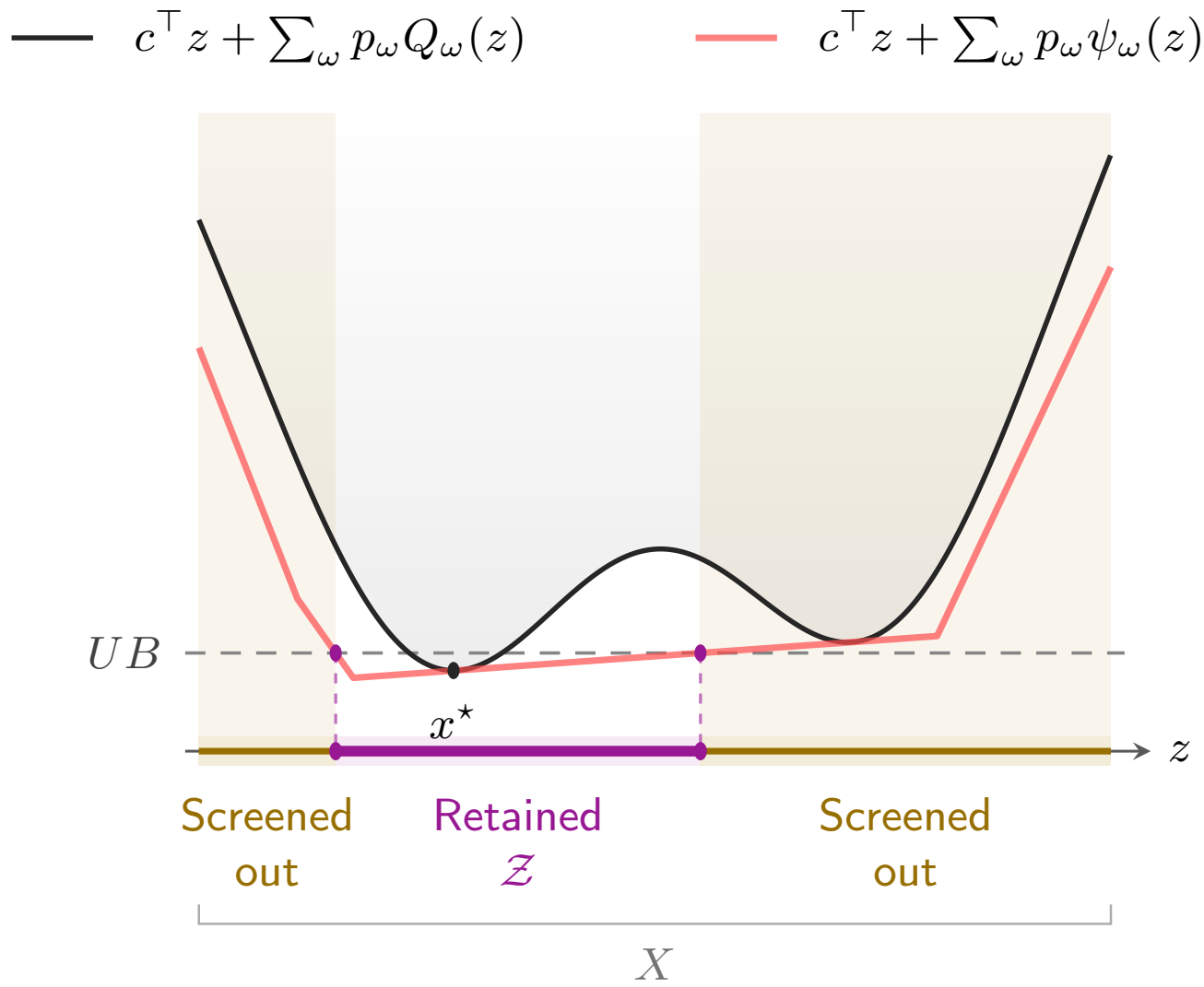
$$\mathcal{Z} \leftarrow \left\{ z \in \mathcal{Z} \;\middle|\; c^\top z + \sum_{\omega \in \Omega} p_\omega \psi_\omega(z) \leq UB \right\}.$$

For every $x^\star \in X^\star \subseteq \mathcal{Z}$,

$$\begin{aligned} c^\top x^\star + \sum_{\omega \in \Omega} p_\omega \psi_\omega(x^\star) &\leq c^\top x^\star + \sum_{\omega \in \Omega} p_\omega Q_\omega(x^\star) \\ &= z^\star \leq UB. \end{aligned}$$

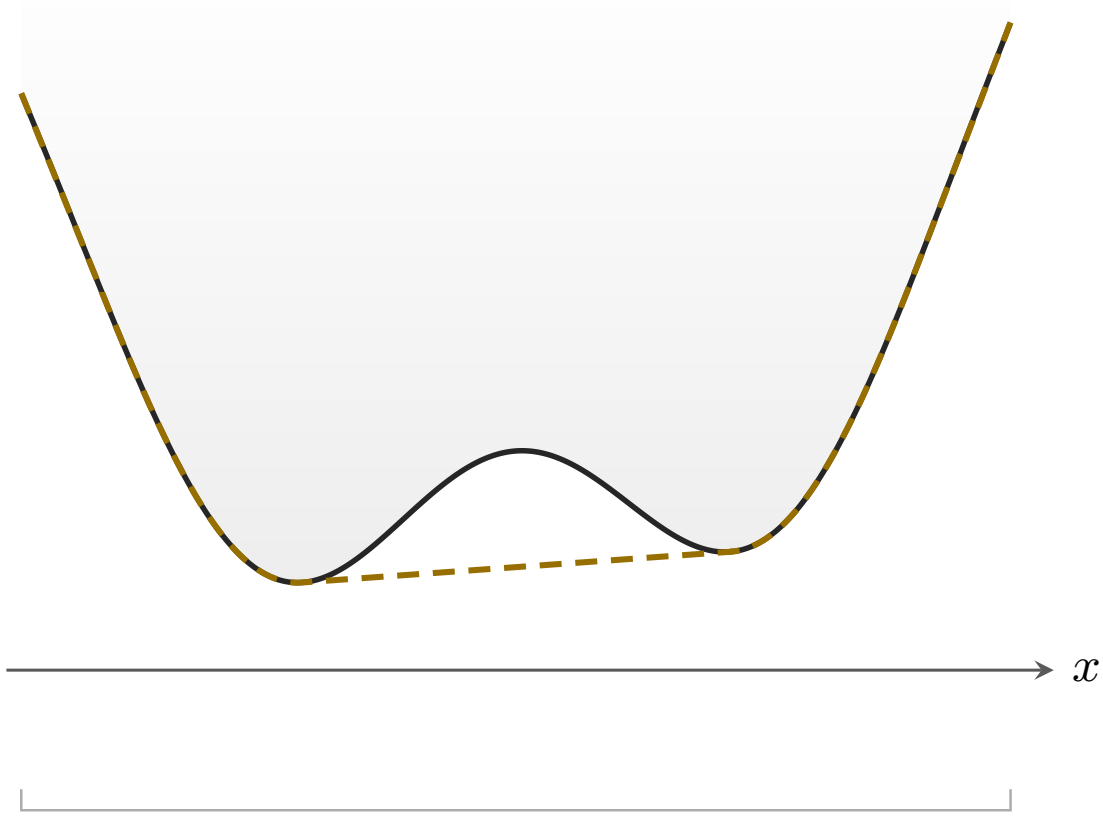
Screening preserves $X^\star \subseteq \mathcal{Z} \subseteq X$.

Screening Geometry



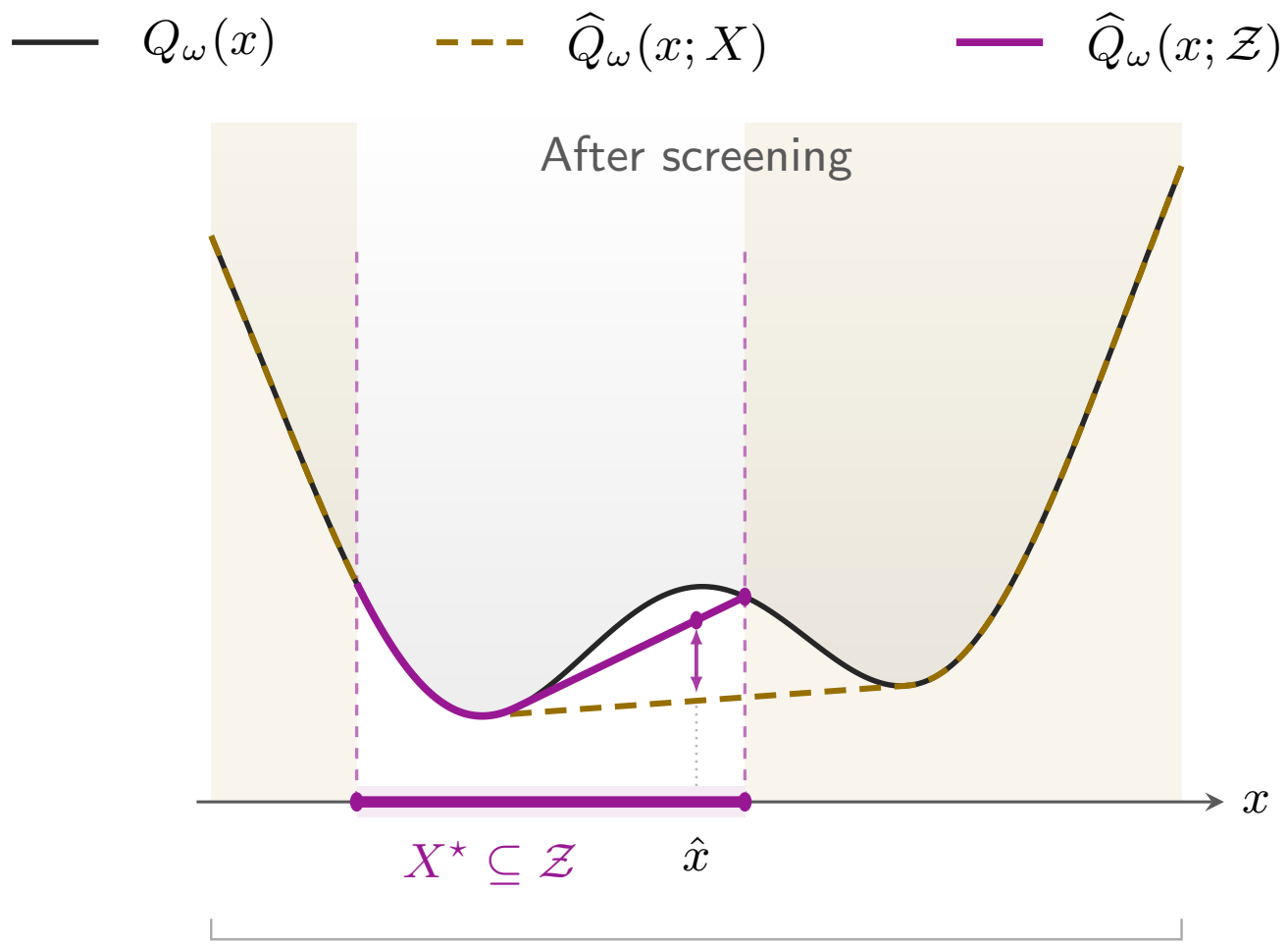
Local Recourse Envelope

$Q_\omega(x)$ $\widehat{Q}_\omega(x; X)$



$$\widehat{Q}_\omega(x; X) \leq Q_\omega(x) \qquad \forall x \in X$$

Local Recourse Envelope



$$\widehat{Q}_\omega(\hat{x}; X) \leq \widehat{Q}_\omega(\hat{x}; \mathcal{Z}) \leq Q_\omega(\hat{x})$$

Main Results

1. **Dantzig–Wolfe cuts**

   With certified pricing, cuts are valid on $\mathcal{Z}$ and satisfy

   $$\phi_\omega^{DW}(\hat{x}) = \widehat{Q}_\omega(\hat{x}; \mathcal{Z}).$$

   Basic optimal dual solutions yield facet-defining cuts.

2. **Local validity**

   Screening preserves $X^\star \subseteq \mathcal{Z} \subseteq X$ and tightens the envelope:

   $$\widehat{Q}_\omega(x; X) \leq \widehat{Q}_\omega(x; \mathcal{Z}) \leq Q_\omega(x), \qquad x \in \mathcal{Z}.$$

3. **Finite convergence**

   For $X \subseteq \{0,1\}^{n_1}$, exact master and recourse solves, together with certified DW pricing, give finite termination at an optimal solution.

Computational Results

Server location and sizing: binary installation, integer assignments

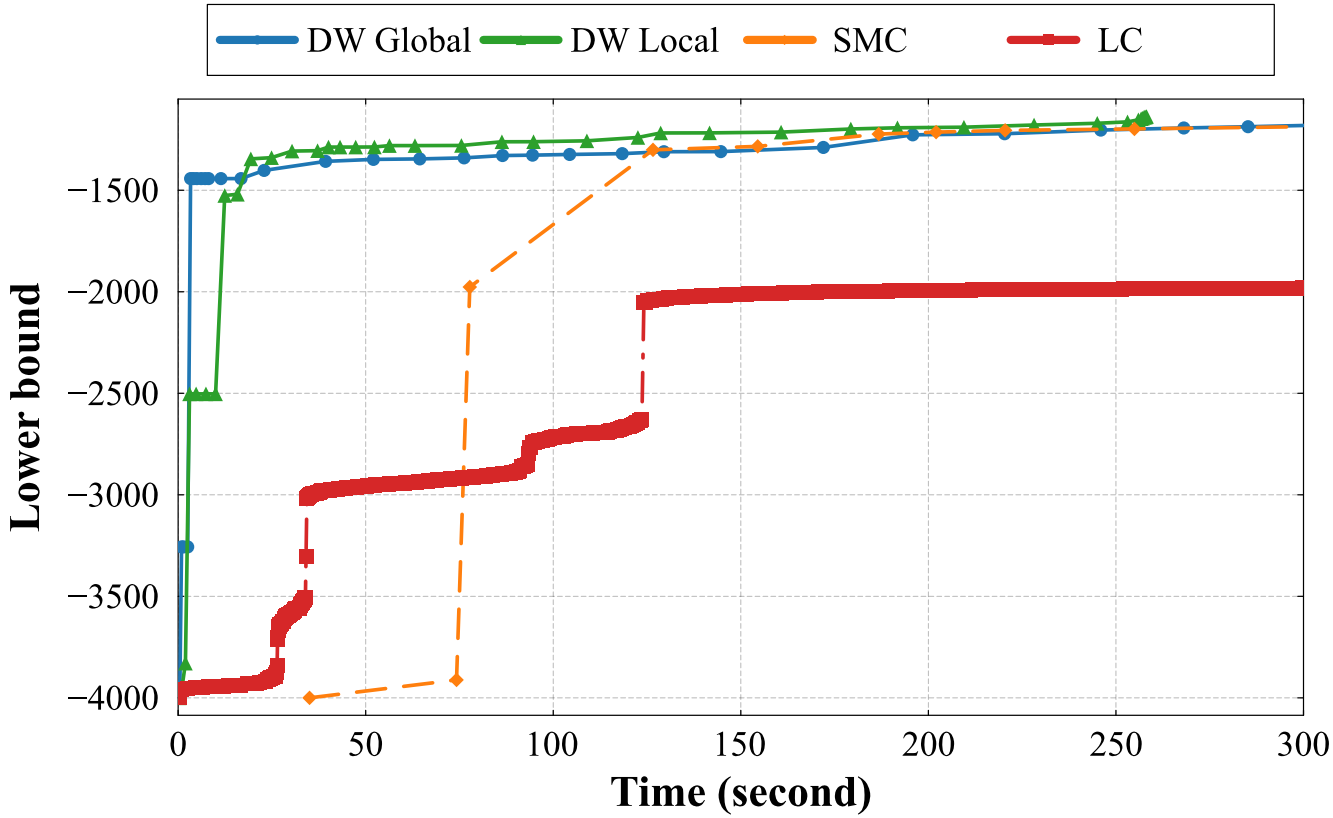
SSLS: Benchmark Comparison & Local Validity

Sites/ customers	LC		SMC		Global DW		Local DW	
	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)
5 / 15	15	0.00	10	0.02	7	0.00	5	0.00
10 / 20	54	0.03	190	0.00	166	0.00	42	0.00
10 / 35	50	0.03	162	0.10	177	0.03	44	0.00
15 / 25	>3600	42.14	>3600	1.25	648	0.10	249	0.00
15 / 30	>3600	42.33	1050	0.00	864	0.03	418	0.00

LC: Lagrangian cuts; SMC: square-minimization cuts.
Local DW includes region-cut filtering.

SSLS: Lower-Bound Trajectories

15 sites, 25 customers, 100 scenarios.

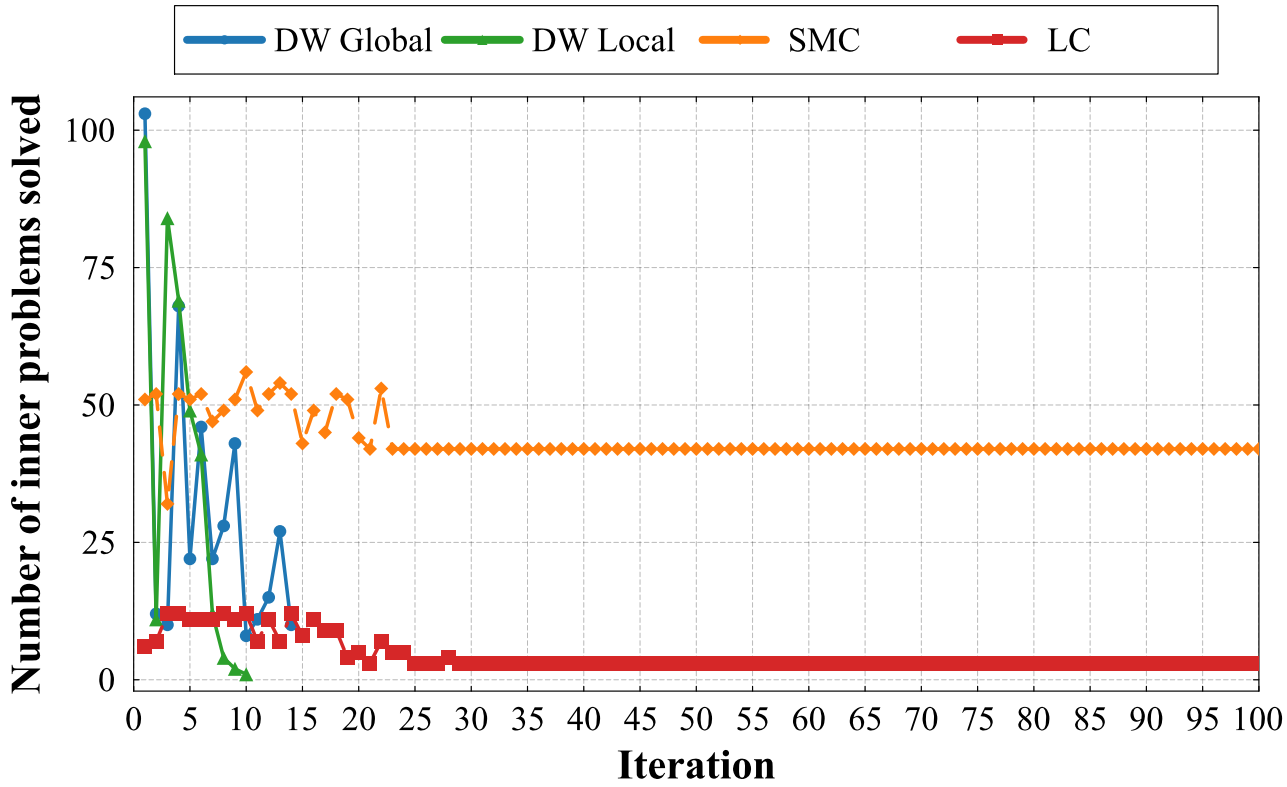


- Local DW: rapid lower-bound improvement.

SSLS: Inner-Problem Workload

15 sites, 25 customers, 100 scenarios.

Per-scenario averages at each outer iteration.



- DW needs additional inner work in later iterations.

Summary

- **Dantzig–Wolfe cuts**
  LP-based generation with pricing, reusable points, and facet selection.

- **Local validity**
  Preserve optimal solutions while tightening the recourse envelope.



Thank you!

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Research Agenda

Optimization under uncertainty

Multistage SP DRO Chance constraints Bilevel

1. Globally Converging Algorithm for Multistage Stochastic Mixed-Integer Programs
   Major revision — *Operations Research*.
2. Contextual Distributionally Robust Chance Constraints: Exact Reformulations and Valid Inequalities
   Submitted — *Operations Research*.
3. Concave-Envelope Cuts for Stochastic Mixed-Integer Bilevel Programs

Research Agenda

Power systems

5. Multi-Period Power System Risk Minimization Under Wildfire Disruptions
   Published — *IEEE Transactions on Power Systems*, 2024.

6. Multistage Stochastic Program for Mitigating Power System Risks under Wildfire Disruptions
   Published — PSCC 2024 / *Electric Power Systems Research*.

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AI and optimization

7. Finite-Sample Probabilistic Safety Certification for AI-Based Grid Operation
   Submitted — *IEEE Transactions on Smart Grid*.

Related Work

Foundation	Connection to this paper
Zou, Ahmed and Sun (2019)	Copied states, Lagrangian cuts, and exactness at binary states.
Chen and Luedtke (2022)	Lagrangian separation with stored recourse points; reuse already appears in this literature.
Chen, Günlük and Lodi (2025)	DW block cuts recover a DW bound in the original variable space.
Füllner, Sun and Rebennack (2024)	Normalization and multiplier selection govern Lagrangian cut geometry.

Here: scenario-wise DW cuts combined with optimality-preserving local regions.

Certified basic LP solutions provide facet selection without an auxiliary selector.

Lagrangian Duality

For the same region $\mathcal{Z}$, define

$$\mathcal{L}_\omega(\lambda; \hat{x}, \mathcal{Z}) = \min_{(z,y)\in\mathcal{S}_\omega(\mathcal{Z})} \{q_\omega^\top y + \lambda^\top(\hat{x} - z)\}.$$

$$\begin{aligned}\max_\lambda \mathcal{L}_\omega(\lambda; \hat{x}, \mathcal{Z}) &= \max_{v,\lambda} \left\{ v + \lambda^\top \hat{x} \;\middle|\; v + \lambda^\top z \le q_\omega^\top y \quad \forall (z,y) \in \mathcal{S}_\omega(\mathcal{Z}) \right\} \\ &= \widehat{Q}_\omega(\hat{x}; \mathcal{Z}).\end{aligned}$$

- Each pricing point supplies one inequality of this dual LP.
- DW and optimized affine LC have the same value on the same domain.
- A smaller validity domain changes the attainable envelope.

Chen and Luedtke (2022), Section 4.2, also reuse stored points across separation calls.

Validity and Tightness

For a certified DW cut generated at $\hat{x} \in \mathcal{Z}$:

$$\phi_\omega^{DW}(x) \le \widehat{Q}_\omega(x; \mathcal{Z}) \le Q_\omega(x), \qquad x \in \mathcal{Z},$$

$$\phi_\omega^{DW}(\hat{x}) = \widehat{Q}_\omega(\hat{x}; \mathcal{Z}).$$

- **Original recourse exactness:** if $\hat{x} \in \operatorname{ext}(\operatorname{conv} \mathcal{Z})$, then $\widehat{Q}_\omega(\hat{x}; \mathcal{Z}) = Q_\omega(\hat{x})$.
- **Facet:** certified pricing, a basic optimal dual solution, and full-dimensional projected domain.
- **Budgeted correction:** validity holds; tightness and the facet property need not hold.

If pricing is interrupted, use an upper bound on the maximum violation in the intercept correction.

Region-Cut Filtering

Write $\psi_\omega(x) = \max_t\{\alpha_{\omega t} + \beta_{\omega t}^\top x\}$. A tuple τ selects one cut per scenario:

$$a_\tau = \sum_\omega p_\omega \alpha_{\omega t_\omega}, \qquad b_\tau = \sum_\omega p_\omega \beta_{\omega t_\omega}.$$

It gives a region cut $(c + b_\tau)^\top z \le UB - a_\tau$.

$$\max_{z \in \text{current region}} \{a_\tau + (c + b_\tau)^\top z - UB\}.$$

- Positive value: retain the cut; it removes part of the current region.
- Nonpositive value: discard the redundant inequality.

State screening removes states.
Region-cut filtering avoids redundant region constraints.

SSLS – Cut Families

100 scenarios; DW uses local validity and region-cut filtering.
Pricing budget grows by one per outer iteration, from 1 to 15.

Sites/customers	Method	Gap (%)	Iter.	Time (s)	MIPs/scenario
15/25	DW	0.00	40	248.97	421
	SMC	1.25	273	–	11,621
	LC	42.14	3,518	–	10,569
15/30	DW	0.00	47	417.61	557
	SMC	0.00	47	1,050.28	2,196
	LC	42.33	3,375	–	10,141

LC and SMC use the level method with relative error 10^{-3} or 200 inner iterations.

DCAP Benchmarks

Capacity acquisition: continuous capacity and binary activation.

200 scenarios; spatial branching is used for the continuous decisions.

DW pricing budget grows by one per outer iteration, from 1 to 15.

Instance	Method	Final gap (%)	Inner MIPs/scenario
233	DW	0.00	407
	SMC	0.44	24,161
	LC	552.96	29,685
342	DW	0.09	1,581
	SMC	0.32	25,636
	LC	126.20	35,828

- LC: Lagrangian cuts; SMC: square-minimization cuts.
- DW gives smaller gaps with fewer inner MIP solves on these instances.

Air-Ground Service Network

592,667 observed orders over seven days in Shenzhen; 20 service zones.

	Air-service share (%)	
Facility limit B	Baseline	Doubled fleet productivity
8	3.96	5.51
20	7.85	11.59

- At each B, the reported facility footprint remains unchanged.
- Higher productivity increases air service within that footprint.

Selected References

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